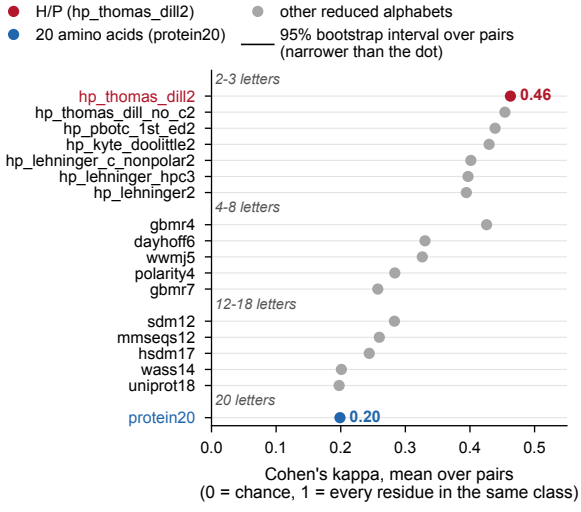
a One Pfam-A 38.2 seed pair at 21% identity: HTH_6 domain (PF01418), RPIR_ECOLI and Y143_HAEIN

H = hydrophobic (ACFILMVWY) P = polar (DEGHKNPQRST) Thomas-Dill split (hp_thomas_dill2)

| same amino acid | same H/P class | longest class-identical run (no gap, same class in both); 12 here, the median of all 37,085 pairs

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RPIR_ECOLI      14 IGLAPYLRMKQEGMTENESRIVEWLLKPGNLSCAPAIKDVAEALAVSEAMIVKVSKLLGFSGFRNLRSALEDYFSQS 90
16 of 77 identical (21%)
Y143_HAEIN       5 GNVLNKIGSLYQSLTKSEKKIADTILRSPDLVSQCSLSEIAKHLQVGEATLVRFCRTIGFKGFSEFKLELSIELATK 81
                   H/P class of the same residues
RPIR_ECOLI        HPHHPHHPHPPPPHPPPPPPPHHPHHHPPPPHPHHPHHPPHHPHHHHPPHHHHPHPPHHPHPPHPPHPPHHPHHPPP
56 of 77 same class (73%)
Y143_HAEIN        PPHHPPHPPPHHPPHPPPPPPHHPPHHPPPPHPHHPHHPPHHPHPPHPPHPHHPHHPPHPHPPHPPHPHPHPHHPP
                                  12 residues
```

b H/P class is kept well above chance (37,085 pairs, 20-30% identity)



c but only in short runs

